

**DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED
BACKEND DEVELOPMENT (25CS1302E)**

Y25 - 2026-2027 Trimester - 4

Project Abstract Submission Form

Sec No-13	Team No-14	Project Title- Warehouse & 3PL Management System
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1. Abstract

- The Warehouse & 3PL Management System is a database-driven web application designed to streamline warehouse operations and third-party logistics (3PL) activities through centralized data management. Traditional warehouse and logistics processes often rely on manual records, disconnected systems, and repetitive data entry, which can lead to inventory inaccuracies, order delays, shipment tracking difficulties, and inefficient resource utilization. The proposed system provides an integrated platform for managing warehouse information, inventory, customers, orders, shipments, and logistics activities in a structured and efficient manner.
- The system uses a relational database to store and organize important operational data, including warehouse details, product information, stock levels, customer records, orders, shipment details, and delivery status. Relationships between these entities are maintained to ensure data consistency, integrity, and efficient retrieval. The backend handles business logic, database operations, and communication between the application and database, while the frontend provides a user-friendly interface for warehouse staff, administrators, and logistics providers.
- The major modules of the system include Warehouse Management, Inventory Management, Order Management, Shipment and Delivery Tracking, Customer/Client Management, and 3PL Administration. The system supports essential database operations such as insertion, updating, deletion, searching, and retrieval of records. By integrating warehouse and logistics processes into a centralized database system, the proposed solution aims to improve inventory visibility, reduce operational errors, enhance shipment coordination, and support efficient decision-making. Overall, the project demonstrates practical applications of database design, relational data management, CRUD operations, and distributed backend development in a real-world logistics environment.

2. Team Details

Roll Number(s)	Name of the Student	Signature
2520080045	U. Sindhuja	
2520080024	P. Bhagya Sri	
2520090039	S. Rishitha Reddy	

3. Frameworks / Platforms / Software Used

- **React.js** – Frontend development and user interface
- **Node.js** – Backend runtime environment
- **Express.js** – Server-side development and REST APIs
- **MySQL** – Relational database management and data storage
- **Nodemailer** – Email notifications and confirmations
- **Razorpay / Stripe** – Online payment integration
- **JavaScript** – Application logic and functionality
- **HTML5 & CSS3** – Web page structure and styling
- **Visual Studio Code** – Development environment

DR.M.Gayatri
Signature

(remarks if any) _____